

SoC May 2026

WEEK 1: Introduction to Reinforcement Learning

Mayuresh Thengzode

25B1816

1.Introduction to RL

Reinforcement Learning (RL) is a branch of machine learning where an agent learns by interacting with an environment. The agent takes actions, receives rewards or penalties, and gradually improves its behaviour to maximize long term rewards. Unlike supervised learning, RL does not rely on labelled data but instead learns through trial and error.

2.Agent and Environment

RL works through continuous interaction between an agent and its environment.

- The agent observes the current state.
- It selects an action.
- The environment responds with a reward and a new state.

3.State and Action

State is the current configuration of the environment and Action is something that agent does in that state.

4.Return

Return represents the total future reward an agent receives. The goal of the agent is to maximise the reward.

5.Policy

A policy defines the behaviour of the agent. It specifies what action should be taken in a given state. Policies may be deterministic or stochastic. The objective of RL is to find an optimal policy that produces the highest expected return.

6.Value Functions

Value functions estimate how good a state or action is.

7.Exploration and Exploitation

- **Exploration:** Trying new actions to discover better strategies.
- **Exploitation:** Using known actions that already provide high rewards.

Effective RL systems balance both to achieve the best long-term performance